

HyperGranular-RAG: Structured Evidence Completion for Compact Multi-Hop Retrieval

Anonymous ACL submission

Abstract

Multi-hop question answering requires both selecting complementary evidence and arranging it for a generator under a limited evidence budget. A relevance ranking can miss a bridge fact, while expansion can displace useful context. We study these decisions through HyperGranular-RAG, a structured evidence-completion layer that groups candidate sentences by embedding geometry, selects query-conditioned facet hyperedges, and inserts a bounded number of candidates after a protected dense prefix. In three closed-candidate evaluations on HotpotQA and MuSiQue, the compact MiniLM implementation improves answer F1 by 0.01140–0.01478 with Qwen2.5-1.5B-Instruct; all three recorded paired bootstrap intervals exclude zero. A component comparison favors facet-conditioned selection over a centroid-only selector. The original system nevertheless trails BGE, and neither a cross-space sidecar nor a BGE-native reconstruction establishes an increment over BGE; generator transfer also remains unresolved. With identical prompt-visible evidence, protected placement outperforms front insertion in the cross-space study. Together, these comparisons support completion for the tested compact backbone and distinguish the benefit of an ordering policy from evidence that its selected candidates improve a stronger retriever.

1 Introduction

Retrieval-augmented question answering separates access to evidence from answer generation. Dense retrieval makes relevance ranking efficient (Karpukhin et al., 2020), but multi-hop questions require several facts to be available together. A passage about one entity may be highly relevant without supplying the bridge needed to connect it to another. HotpotQA and MuSiQue provide complementary settings for examining this difficulty (Yang et al., 2018; Trivedi et al., 2022).

The challenge becomes sharper under a fixed evidence budget. Adding a candidate does not enlarge a Top-20 context indefinitely: it can promote an already retrieved sentence or displace a different one. Moreover, a generator does not use all retrieved information equally. Evidence position affects context use (Liu et al., 2024), so improved retrieval coverage need not produce an improved answer. A useful completion method must balance candidate relevance, complementarity, and the cost of disturbing the original ranking.

We investigate a structured completion layer, HyperGranular-RAG, that preserves the retrieval backbone. It organizes the available sentence embeddings into local granular balls, uses query-term facets to relate seed balls to candidate expansion balls, and places at most four candidates after ten protected entries. *Completion* describes an intervention on the evidence list; it is not a guarantee that all necessary facts have been recovered. The compact selector is a static, seed-relative heuristic, not a dynamically optimized coverage objective.

Our central question is how evidence selection and arrangement contribute to end-to-end answer quality. Full-versus-Dense comparisons evaluate the completion pipeline, and Full-versus-NoFacet tests its specified selector. The cross-space Protected-versus-Unprotected comparison holds visible evidence fixed and tests its ordering. Comparisons with BGE and one additional generator then examine the boundaries of these findings. Each contrast addresses a different claim: a better ordering need not imply that the selected candidates improve the original ranking.

The results reveal a consistent but bounded opportunity. The compact system improves over MiniLM Dense in three evaluations, and its facet-conditioned selector exceeds the specified centroid-only comparator. The original compact system remains below BGE. Neither transferring its candidates to a BGE ranking nor reconstructing the

geometry in BGE space establishes additional answer quality over BGE. Thus, the positive compact result and the unresolved strong-backbone extensions are part of the same empirical account.

We make three contributions:

- We specify a bounded, geometry-based completion layer over a dense ranking, making its evidence-selection and placement decisions explicit.
- We report repeated compact-backbone improvements and component, generator, and strong-backbone comparisons, retaining their distinct data boundaries and dataset-specific evaluation definitions.
- We distinguish selection from arrangement through a verified fixed-visible-evidence placement comparison and supporting-evidence turnover analysis, clarifying what the observed gains can be attributed to.

2 Related Work

Retrieving connected evidence. Dense passage retrieval learns representations for relevance ranking (Karpukhin et al., 2020). Multi-hop systems can instead retrieve iteratively: MDR conditions later retrieval on earlier evidence (Xiong et al., 2021), whereas IRCOT interleaves retrieval and reasoning (Trivedi et al., 2023). Our intervention is a single completion of a fixed candidate ranking. It does not search an external corpus or rewrite queries using generated reasoning.

Granular-ball and hypergraph learning. Granular-ball representations precede this study (Xia et al., 2019). SAGHL constructs hierarchical hyperedges from graph topology (Ni et al., 2026); the MGHRL preprint combines topological ball decomposition with a trained multi-granularity network (Zhao et al., 2026). Granular-ball divergence also supports fuzzy rough hypergraph feature selection (Li et al., 2026). Thus, combining granular balls and hypergraphs is not itself our novelty claim. Our evaluated groups organize sentence embeddings for inference-time selection, rather than graph classification or feature reduction. These task differences still leave the need to compare grouping rules under matched QA budgets.

Hypergraph and cross-granularity RAG. RAPTOR and HippoRAG organize evidence using hierarchies and knowledge graphs (Sarathi et al., 2024; Gutiérrez et al., 2024). The closest task-level neighbor is HGRAG (Wang et al., AAAI 2026): entities are nodes, passages are hyperedges, and query-conditioned diffusion combines entity and passage similarity (Wang et al., 2026). HyperGraphRAG retrieves extracted n-ary facts (Luo et al., 2025), while Hyper-RAG combines extracted low- and high-order relations with corpus text (Feng et al., 2026). *HyperGranular-RAG* denotes our method throughout; it is distinct from Wang et al.’s HGRAG. Our facet hyperedges are lexical selection relations between local sentence groups, without LLM fact extraction or learned graph propagation. This is an operational distinction, not evidence of superiority over those systems. Their corpus, reader and context budgets differ from ours; their published scores are not entries in our empirical comparison.

Diversity, coverage, and context use. MMR balances relevance and redundancy (Carbonell and Goldstein, 1998). Coverage and diversity can also be represented by explicit submodular objectives (Lin and Bilmes, 2011). Such methods are important alternative explanations for a completion gain; the historical study did not run budget-matched versions of them. We therefore compare the facet selector only with its actual centroid-only control. Long-context position effects (Liu et al., 2024) motivate protection, but do not establish that a particular protected-prefix length is optimal.

3 Method

3.1 Evidence Budget and Local Organization

For question q , let U_q be its benchmark-supplied candidate sentence units. The retriever embeds the query and each title–sentence pair, normalizes the embeddings, and scores

$$s(q, u) = \hat{e}_q^\top \hat{e}_u. \quad (1)$$

Descending score and ascending unit identity determine the dense order D_q . Every arm returns $K_q = \min(20, |U_q|)$ unique units. The compact variant uses the same MiniLM space for ranking and organization.

For a nonempty group B , define

$$\begin{aligned} c_B &= \text{norm} \left(\sum_{u \in B} \hat{e}_u \right), \\ r_B &= \max_{u \in B} (1 - \hat{e}_u^\top c_B). \end{aligned} \quad (2)$$

The root contains all candidates, not only Dense Top-20. The compact implementation attempts a split when depth is below six, $|B| \geq 4$, and $|B| > 3$ or $r_B > 0.78$. The radius condition is redundant under these size settings. A farthest-from-center seed and its least-similar partner partition the group by cosine similarity. Both children must contain at least two units; otherwise the group is retained. Consequently, three is not a strict maximum leaf size. Geometry determines assignments, but radius does not independently trigger splits in this compact configuration.

The resulting groups are termed granular balls, following the broader idea of granular representations (Xia et al., 2019). This terminology does not assert that these groups outperform ordinary clustering.

3.2 Static Facet-Conditioned Selection

Let T_q be the content terms of the query under the fixed ASCII-alphanumeric tokenizer and stop-word list. A ball’s facets are the intersection of T_q with the union of terms in its titles and sentences. The two balls with highest query–centroid score form the seed set. An expansion hyperedge relates these seeds to a non-seed ball contributing at least one additional query term.

For candidate B , let n_B and t_B denote seed-relative new and total facet counts; $m = |T_q|$, $b_B = \hat{e}_q^\top c_B$, a_B is maximum seed-centroid similarity, and ρ_B is the fraction of its facets already in the seeds. The compact score is

$$\begin{aligned} h_B &= .45 \frac{n_B}{\max(m, 1)} + .20 \frac{t_B}{\max(m, 1)} + .20 b_B \\ &\quad + .15(1 - a_B) - .20 \rho_B - .05|B|/8. \end{aligned} \quad (3)$$

Eligibility additionally requires $|B| \leq 8$, $b_B \geq .12$ or $a_B \geq .05$, at most six units per new term, $\rho_B \leq .90$, and $h_B \geq .10$. At most two expansion balls survive descending h_B and stable edge-identity ordering. Units inside them retain their dense-score order.

All scores are computed once relative to the seeds. Accepting one ball does not change another’s score. Two accepted balls may therefore

contribute the same new term. Nor does selecting a ball imply that every facet in its union occurs in the selected sentence or survives prompt truncation.

3.3 Protected Insertion

The compact policy preserves the first $\min(10, K_q)$ dense units, traverses the candidate order, and inserts at most four units absent from that prefix with score at least 0.1957079917192459. This development-calibrated 25th-percentile floor gives the policy its internal name, static-q25; it is not re-estimated per confirmation query. Remaining dense units are appended in order, with stable deduplication and truncation to K_q .

A placed unit can already belong to Dense ranks 11–20. We distinguish such a promotion from a genuinely new unit outside Dense Top-20. Equal K_q also does not imply equal token count: all methods share a 4,096-token input cap, while actual evidence length varies. Figure 1 separates selection, placement, and prompting.

3.4 Two Extensions around BGE

The cross-space *sidecar* keeps BGE’s primary ranking and uses the frozen MiniLM structure to propose candidates outside BGE Top-20. Its protected and unprotected arms use the same insertion candidates but place them after the prefix or at the front, respectively.

The *native* variant reconstructs groups and relative scores in BGE space. Development selected one of 16 configurations, with leaf target six, four insertion slots, unit-score percentile floor .50, and broad relative facet gates. The split depth, balance gate, score normalization, and candidate score differ from the compact method (Appendix A). This is a related implementation, not an encoder-only substitution.

4 Experimental Design

4.1 Questions and Comparisons

We use HotpotQA training distractor candidates and the answerable MuSiQue training candidate sets (Yang et al., 2018; Trivedi et al., 2022). The compact evaluations contain 1,000 HotpotQA questions, 3,000 MuSiQue questions, and a separate joint component set of 1,000 and 1,500 questions. The sidecar and native confirmation studies each use another 1,000/1,500 set. Their frozen identity manifests record exclusion of the preceding evaluation boundaries. These are project-held-out

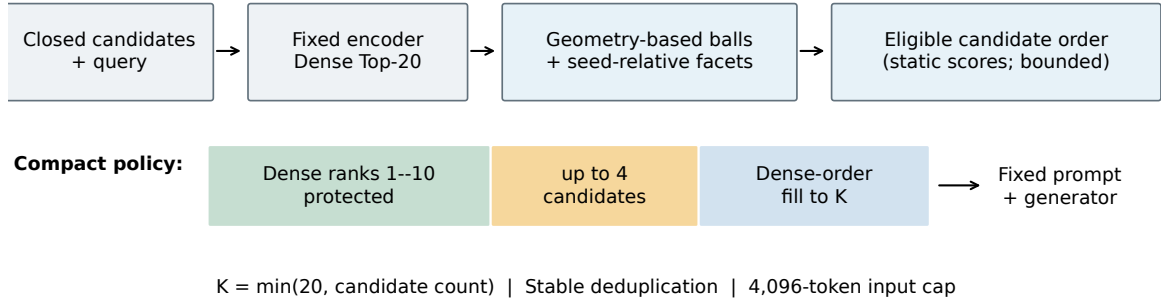


Figure 1: The compact pipeline. Candidate groups cover the full closed candidate set. Selection is static relative to the seeds; the protected prefix, candidate placement, dense-order fill, and subsequent token cap are separate operations. No answer label enters retrieval or prompting.

subsets of training releases, not official hidden test submissions.

The joint component study includes MiniLM Dense, BM25, a Dense–BM25 hybrid, BGE large-en-v1.5, FULL, NoFacet, and NoProtection. NoFacet retains compact ball geometry but uses centroid-only expansion. NoProtection changes the placement policy. Neither is a flat-unit or budget-matched MMR control. Native development uses 500 HotpotQA and 750 MuSiQue questions, separate from confirmation. Its small positive development difference selects a configuration; it is not pooled with confirmation.

4.2 Generation and Metrics

The principal generator is Qwen2.5-1.5B-Instruct FP16 on an RTX 4060 Laptop 8GB. A fixed short-answer prompt requests an answer from the evidence or UNKNOWN. Generation is greedy, batch size one, with at most 32 new tokens and a 4,096-token input cap. A transfer study uses the same initial rankings with Gemma 4 E2B in the official mobile-QAT deployment format. That comparison concerns these deployment configurations, not pure model architecture or device-independent efficiency.

We report mean answer F1 and exact match (EM). Complete recall (CR@20) records whether all annotated supporting evidence is covered; evidence recall (ER@20) records the covered fraction. HotpotQA has sentence-level support labels, whereas MuSiQue uses supporting paragraphs. MuSiQue coverage therefore must not be described as sentence-chain recall.

We use each dataset’s pinned official answer functions. HotpotQA applies its yes/no/noanswer

exception and assigns zero F1 when normalized tokens have no overlap; MuSiQue maximizes over aliases and treats two empty token lists as a match. A retrospective scoring reconciliation preserves all confirmation and generator-transfer scores and all method differences; a common development-score correction did not trigger configuration reselection (Appendix A). These are answer metrics on project subsets, not official leaderboard or joint answer-and-support evaluation.

4.3 Estimates and Uncertainty

For paired per-query scores, the target is the mean within-query difference. Joint studies average the two dataset means with equal weight. Their 10,000 bootstrap draws resample queries within each dataset, preserving method pairing. We reconstructed the original intervals and reused the same draws for canonical scoring; all confirmation intervals are unchanged. The intervals describe the specified query-sampling model, not uncertainty over model training or alternative development searches.

Statistical tests require a null distribution and suitable sampling assumptions (Dror et al., 2018). Some historical decision files call an uncentered bootstrap nonpositive-tail fraction a one-sided p-value and apply Holm adjustment. Its general null calibration was not established. We preserve the recorded decisions for provenance, but base the present description on effect estimates and intervals, without claiming a newly validated family-wise test. Intervals crossing zero are unresolved, not equivalent. Shared source documents can induce dependencies not captured by query-level resampling.

5 Results

5.1 Does Completion Help Compact Retrieval?

Yes, within the tested compact configuration. Table 1 shows positive effects on the initial HotpotQA and MuSiQue sets and the separate component set. Their recorded 95% intervals exclude zero, with absolute F1 differences between .01140 and .01478. The initial HotpotQA lower bound is close to zero, so the magnitude and uncertainty are more informative than a binary success label.

Retrieval coverage also increases: CR@20 rises from .737 to .798 on HotpotQA and from .589 to .650 on MuSiQue. These are different support granularities, and neither change by itself measures answer improvement. The corresponding answer gains supply the end-to-end evidence.

5.2 Selection and Placement Contrasts

On the joint compact set, FULL minus NoFacet is +0.01336 with interval [0.00341, 0.02343]. This favors the particular facet-conditioned selector over centroid-only expansion. It does not show that hyperedges are necessary, or that higher-order structure explains more than a simpler coverage heuristic. FULL minus NoProtection is +0.00354 [−0.00675, 0.01389], leaving the independent benefit of protection unresolved in that compact setting.

In the sidecar study, Protected minus Unprotected is +0.01122 [0.00129, 0.02104]. Protected insertion is better than the tested front insertion policy at fixed visible evidence. Protected minus BGE remains unresolved: the two comparisons test ordering and BGE-relative answer quality, respectively. The native facet and placement contrasts also cross zero (Figure 3).

5.3 Does the Benefit Extend to Strong Retrieval?

The original compact FULL is below BGE on the same component set: −0.03998 [−0.05393, −0.02621]. BM25 and the hybrid are additional comparators, with FULL differences of −0.00755 [−0.02259, 0.00773] and −0.00320 [−0.01591, 0.00942], respectively. Neither interval resolves a difference.

The two BGE extensions address more specific alternatives. The sidecar adds candidates from the existing compact structure; its Protected-minus-BGE effect is −0.00256 [−0.00998, 0.00458]. Re-

building the geometry natively in BGE yields −0.00305 [−0.00688, 0.00063]. Both point estimates are slightly negative, and both intervals cross zero. These findings do not establish complementarity with BGE, but they are not equivalence tests. Because the configurations and query sets differ, the three BGE comparisons are not a causal dose–response curve for retriever strength. Figure 2 places these distinct effects and intervals on a common display scale.

5.4 Does the Result Transfer to Another Generator?

With the fixed Gemma mobile-QAT configuration, the equal-weight FULL-minus-Dense effect is +0.00516 [−0.00262, 0.01295]. This result does not establish generator transfer. The rankings are reused from the initial studies, so this is an intervention on the tested generator deployment, not an additional independent retrieval sample. Differences in format and runtime prevent interpretation as a controlled architecture comparison between Qwen and Gemma.

6 Analysis and Implications

6.1 Adding Evidence Can Also Displace It

The post-decision sidecar audit counts 21 added and 29 displaced annotated supporting units in HotpotQA, and 59 added versus 50 displaced in MuSiQue. The corresponding native counts are 3/2 and 16/15. Thus, native completion has a net +1 supporting unit in each dataset while both answer-F1 point estimates are negative. These totals concern annotated evidence identities at each dataset’s support granularity, not complete reasoning chains or a causal effect per inserted unit.

A gain can fail to reach the answer because retained evidence is redundant, an essential fact is displaced, or the generator fails to use the context. The aggregate counts do not distinguish these mechanisms. They demonstrate why selection, final ranking, prompt-visible evidence, and answer quality should be measured separately.

6.2 What Geometry Does and Does Not Establish

Mean-cosine compactness equals the normalized length of the sum of unit embeddings. Its weighted child average cannot decrease under any partition (Appendix B). Thus, increased compactness alone

Evaluation set	Queries	Dense F1	Full F1	$\Delta F1$	95% interval
HotpotQA	1,000	.42150	.43628	+0.01478	[0.00020, 0.02988]
MuSiQue	3,000	.13595	.14735	+0.01140	[0.00450, 0.01835]
Joint component	2,500	.30446	.31803	+0.01357	[0.00491, 0.02233]

Table 1: Compact MiniLM with Qwen. Scores are on the 0–1 scale. The joint row is dataset-equal-weighted, not a pooled average of all 2,500 questions. Dataset-canonical rescoring reproduces every displayed answer score and interval from the frozen predictions.

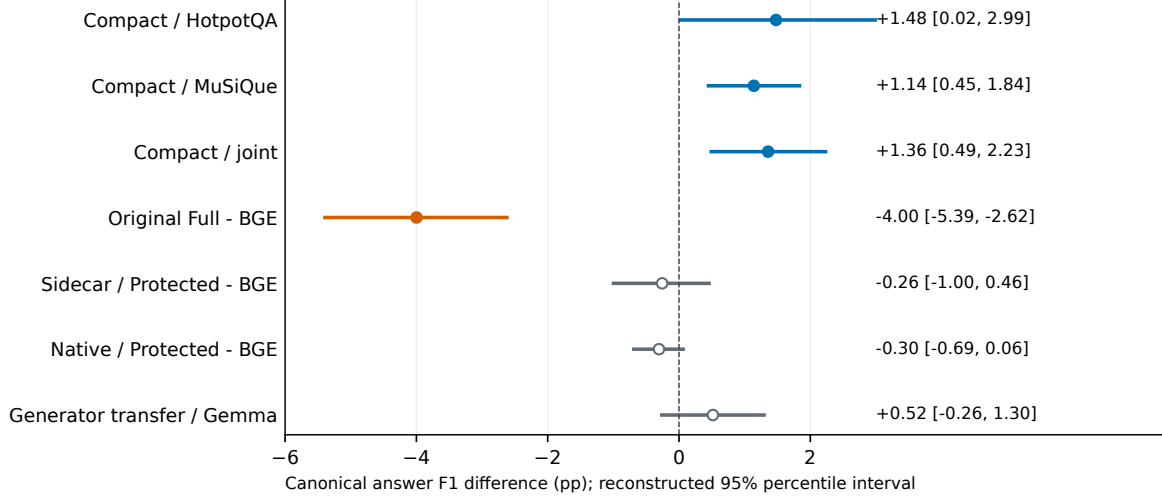


Figure 2: Paired answer-F1 effects in percentage points (one point is .01 on the score scale). Each interval is the recorded 95% percentile bootstrap interval for that row’s comparison, reproduced under canonical scoring; rows are not pooled. Open markers denote intervals crossing zero.

does not establish a meaningful semantic split or explain the observed answer gain.

6.3 Limits of Structural Attribution

The tested pipeline can improve a compact backbone without demonstrating that every named component is necessary. Ordinary clusters, MMR, and sentence-level coverage remain plausible alternatives. A decisive comparison would match candidate access, intervention budget, group count, and generator context. The absence of that comparison narrows the present contribution to a complete pipeline and specific component contrasts.

Table 2 separates the conclusions available from the completed comparisons from the alternatives they do not exclude. A larger Dense list and a shared token-cap comparison would additionally be needed to establish a quality–cost advantage; the present Top-20 results do not establish one.

6.4 Arranging a Fixed Evidence Set

For the cross-space contrast, reconstruction covers all 2,500 questions: Protected and Unprotected have identical final members and prompt-visible ti-

Completed contrast	Scope of the evidence
Compact Full vs Dense	Pipeline gain; grouping vs simple coverage remains open
Compact Full vs NoFacet	Specified selector gain; not a matched MMR comparison
Protected vs front insertion	Ordering effect with fixed visible evidence
Full / extensions vs BGE	Original Full worse; extension gains unresolved

Table 2: Attribution boundaries. No unexecuted baseline is assigned a score, and no comparison to an external hypergraph RAG system is implied.

ties and sentence texts. Each of the 5,000 serialized prompts matches its own recorded identity and token count. There is no evidence dropping or partial truncation. The effect retains all original queries as its denominator, including those with unchanged ordering. It estimates an ordering/placement-policy effect at fixed visible content, without identifying an internal attention mechanism. The separate answer-quality comparison with BGE remains unresolved.

The twenty-entry budget is distinct from token capacity. Protected and Unprotected have

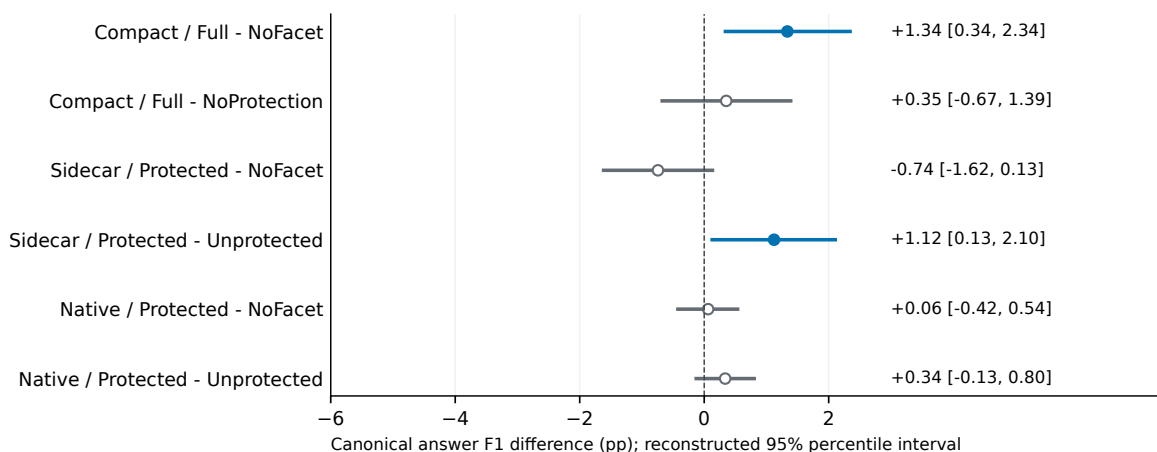


Figure 3: Component contrasts under dataset-canonical answer scoring. NoFacet is a specific historical comparator; native NoFacet also changes the candidate-score coefficients, so that row is a pipeline contrast rather than a pure removal of one coefficient. The sidecar placement comparison fixes actual visible evidence; this verification is not extrapolated to other variants.

identical per-query token counts: HotpotQA median/p95/max are 923.5/1,156/1,907 and MuSiQue 884/1,186.1/2,881. None reaches 4,096. The observed evidence displacement is due to list membership, not exhaustion of the context-token limit.

7 Conclusion

Structured evidence completion improves the tested compact MiniLM pipeline on three closed-candidate multi-hop QA evaluations. A facet-conditioned selector also exceeds its specified compact centroid-only comparator. The original system remains below BGE, while cross-space and native BGE extensions do not establish an incremental answer-quality benefit. At fixed visible evidence, protected placement improves on front insertion in the sidecar study; the comparison with BGE has not established an answer-quality gain. Evidence selection and arrangement therefore require distinct comparisons, with answer quality as the endpoint and coverage and displacement as complementary descriptions of the intervention.

8 Limitations

All answer-quality results use closed benchmark candidate sets. They do not measure corpus reachability, ANN recall, full-wiki indexing, or open-domain noise. The positive compact evidence uses one retriever, one principal generator, one short-answer prompt, and Top-20. The additional generator and BGE extensions are unresolved rather than demonstrations of broad robustness.

The radius gate is inactive in the compact split rule, and the facet selector does not dynamically recompute coverage. The native NoFacet score changes multiple coefficients; its effect is not a pure component ablation. No fair ordinary-clustering or generic coverage comparison has been completed.

The calibration of archived bootstrap-tail tests remains unestablished; scoring reconciliation does not validate those tests. Query-disjoint samples are not necessarily independent at the document or pre-training level. Dataset training releases may have appeared in generator pretraining. Post-decision mechanism summaries and representative cases do not identify causal mediation, and finite resource-based sample sizes are not a power guarantee.

Ethical and Reproducibility Considerations

This study uses existing benchmark releases and model artifacts; it introduces no new participant study. Negative and unresolved results are retained. Source code, configurations, derived predictions, and result manifests exist in the research repository. Third-party datasets and weights remain subject to their original terms. An anonymous distribution and author-approved licensing statement are required before submission. AI-assisted coding, analysis preparation, and writing were used; responsibility for the submitted work remains with the human authors.

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A Historical Implementation and Provenance

The compact method uses all-MiniLM-L6-v2; the strong ranking uses BAAI/bge-large-en-v1.5. Native C10 uses depth limit eight, minimum child fraction .20, and compactness-gain threshold 10^{-10} . Its two seed balls are ranked by query-score percentile,

density percentile, then identity. Broad gates require query-score percentile at least .35 and density percentile at least .25. Eligible non-seed balls must contribute a seed-relative new facet; at most four balls supply candidates. Only units outside BGE Top-20 with unit-score percentile at least .50 are eligible. Candidate scoring is

$$h_u^{\text{native}} = .45p_u + .25p_{b(u)} + .15p_{\rho(u)} + .10p_{C(u)} + .05 \min(n_{b(u)}, 3)/3, \quad (4)$$

where the p terms are within-query empirical percentiles of unit relevance, ball relevance, density, and compactness. Native NoFacet replaces this by $.50p_u + .30p_{b(u)} + .10p_{\rho(u)} + .10p_{C(u)}$ and removes the new-facet eligibility requirement. The comparison therefore changes both gating and weights.

The paper’s scientific names map to the frozen repository records as follows: initial HotpotQA/MuSiQue to Stage4E/4F, generator transfer to Stage4G, joint components to Stage4H, cross-space sidecar to Stage4I, and native confirmation to Stage5A. Aggregate summaries, implementation configurations, and the manuscript evidence ledger provide the exact artifact paths. The later Stage6 development outputs, engineering repairs, and synthetic v2 prototype are excluded from these empirical results.

Scope of the follow-up proposal. A subsequent bounded comparison proposal retained matched MMR, selected-sentence coverage, ordinary grouping, larger Dense contexts, and a shared strong retrieval score. It was not executed because prior exposure of the available confirmation set could not be established. No substitute confirmation set was selected. This eligibility limitation supplies neither a new negative algorithm result nor evidence against the historical compact gains.

Scoring reconciliation and verification scope. The retrospective reconciliation reconstructs legacy scores from stored predictions and applies dataset-canonical scoring with the original resampling draws. It covers 74,750 prediction records across method arms, not 74,750 independent questions. All 53,500 confirmation and generator-transfer records retain their scores. Native development has 17 affected predictions on one question, equally across all 17 methods, leaving every method difference unchanged; no configuration was reselected. Retrieval and generation were not rerun, and this

additional check is not new independent confirmation. Verification records describe their original scope; subsequent algorithm repairs do not retroactively certify those results.

Complexity and measured cost. With embeddings available, recursive two-seed assignment costs $O(H|U_q|d)$, scoring and sorting cost $O(|U_q| \log |U_q| + |\mathcal{B}_q|d + |\mathcal{B}_q| \log |\mathcal{B}_q|)$, and working memory is $O(|U_q|d + |U_q| + |\mathcal{B}_q|)$, excluding tokenization and model inference. The native confirmation records 8.544 s retrieval reconstruction, 887.650 s BGE generation, 884.198 s protected-native generation, and 3.592 GiB peak GPU memory. These are transaction-specific measurements, not corpus-index costs or a controlled speed advantage. Generation seconds accumulate checkpoint calls across resumed processes; they are not the wall time of the final process.

B Geometric Clarification

For unit vectors in nonempty B , mean-cosine compactness is $C(B) = \|\sum_{i \in B} x_i\|/|B|$. The triangle inequality gives $\sum_j |B_j|C(B_j)/|B| \geq C(B)$ for any partition into nonempty groups. This elementary property is explanatory background, not a new theoretical contribution or an answer-F1 guarantee. At zero sum, the scalar extends to zero but a normalized center requires an implementation convention.

C Reproduction Details

Retrieval and units. The compact encoder is all-MiniLM-L6-v2, with attention-mask mean pooling, L2 normalization, and a 192-token encoder limit. BGE uses bge-large-en-v1.5, CLS pooling, L2 normalization, and a 512-token encoder limit. Its query instruction is “Represent this sentence for searching relevant passages:” followed by the question. Document inputs concatenate the title and sentence. HotpotQA preserves the supplied sentence boundaries; MuSiQue splits whitespace-normalized paragraphs with the frozen punctuation-and-capitalization rule. Neither splitter is changed in this review. The reproducibility configuration pins the encoder revisions and unit IDs.

Lexical baselines. BM25 uses lowercase alphanumeric tokens, $k_1 = 1.5$, and $b = 0.75$ within each query’s candidate set. Its inverse document

frequency is $\log(1 + (N - df + 0.5)/(df + 0.5))$. The hybrid baseline averages query-local min-max-normalized dense and BM25 scores with equal weights; a constant score vector maps to zero. Unit ID breaks score ties. These are local closed-candidate baselines, not corpus-scale retrieval systems.

Generation and budgets. The principal reader is Qwen2.5-1.5B-Instruct in FP16, using greedy decoding, one beam, batch size one, and at most 32 generated tokens. The full serialized input, including the chat template, is capped at 4,096 tokens. The system instruction is: “Answer the question using only the provided evidence. Return only the shortest final answer. If the evidence is insufficient, return UNKNOWN.” Evidence is rendered in rank order as numbered title-sentence entries, followed by the question and “Final answer:”. The Qwen runner sets CPU/CUDA seeds to zero, uses one CPU thread, and enables deterministic algorithms. The renderer admits whole units in order until the cap; only an overlong first unit permits partial truncation. The compact policy protects ten dense positions and places at most four candidates under a final budget $K = \min(20, |U_q|)$. Its frozen q25 floor is 0.1957079917192459. BGE-native has its own recorded eligibility policy; it is not retrospectively identified with compact static-q25.

Scoring and interpretation. The reanalysis fixes the original predictions and reference channel. It executes the pure answer-metric functions from the pinned official HotpotQA and MuSiQue implementations, with maximum-over-alias scoring only where the dataset permits aliases. HotpotQA assigns zero token F1 to mismatched yes/no/noanswer responses, and zero F1 when both normalized strings are empty. MuSiQue’s official token F1 assigns one to the latter case and does not inherit HotpotQA’s categorical exception. The archived per-query scores, means, and paired percentile intervals are first independently reconstructed. Canonical scores then reuse the same 10,000 bootstrap draws. This isolates scorer migration from statistical redesign: no new confirmatory test, model selection, or independent confirmation is claimed. Supporting-evidence coverage retains the dataset’s original sentence or paragraph granularity and is not a complete-chain recovery metric.